
PROFORMA FOR THE APPROVAL OF THE THIRD SEMESTER MINI PROJECT

Mini Project Proposal No : _____
(Filled by the Department)

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Comments/Suggestions :

First Review Comments/Suggestions :

<u>Second Review Comments/Suggestions :</u>

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1. Introduction

Customer support is essential for maintaining customer satisfaction and business reputation. Organizations receive large volumes of support tickets daily, making manual ticket analysis, categorization, prioritization, and solution identification time-consuming and error-prone.

CompassIQ is an AI-powered customer support ticket intelligence system that automates ticket management using Natural Language Processing (NLP) and Machine Learning. It analyzes customer complaints, predicts ticket categories and priority levels, assigns them to suitable departments, and recommends solutions by finding similar previously resolved cases.

By combining AI with modern web technologies, CompassIQ helps reduce response time, improve support efficiency, and enhance customer experience.

2. Motivation

Modern customer support centers often struggle with increasing volumes of support tickets while maintaining fast and accurate responses. Traditional ticket management systems primarily depend on manual categorization and routing, leading to delayed responses, incorrect department assignments, inconsistent prioritization, and repeated effort in resolving similar issues. As organizations grow, these limitations reduce customer satisfaction and increase operational costs.

The motivation behind CompassIQ is to develop an intelligent support ticket management system that utilizes Natural Language Processing and Machine Learning to automate ticket analysis and routing. Rather than depending entirely on manual decision-making, the system provides data-driven predictions by learning from historical customer support data. The project also demonstrates the practical application of Machine Learning, semantic search, information retrieval, database management, and web application development to solve a real-world business problem.

3. Objective / Problem Statement

Problem Statement

Organizations handle numerous customer support requests daily, making manual ticket categorization, priority assignment, and solution searching time-consuming and inefficient. This can lead to delayed responses, incorrect routing, and reduced support productivity.

Objectives

The proposed system aims to:

- Develop an AI-powered customer support ticket management platform.
- Automatically classify tickets and predict their priority using Machine Learning.
- Route tickets to the appropriate support department.
- Retrieve similar resolved tickets and recommend suitable solutions.
- Provide analytics and AI performance monitoring for administrators.
- Apply NLP and Machine Learning techniques to automate customer support processes.

4. Functionalities

The proposed system consists of four modules: Customer Module, Machine Learning Intelligence Module, Support Agent Module, and Admin Module.

1. **Customer Module:** Allows users to register, submit support tickets with issue details, track ticket status, view AI-recommended solutions, receive agent responses, and close resolved tickets.
2. **Machine Learning Intelligence Module:** Uses NLP techniques to analyze ticket content, predict categories and priorities using Machine Learning models, retrieve similar resolved tickets using semantic similarity, and recommend suitable solutions from past cases.

3. **Support Agent Module:** Enables agents to manage assigned tickets, view AI predictions and recommendations, respond to customers, update status, add resolution notes, and improve the knowledge base.
4. **Admin Module:** Provides a dashboard to manage users, agents, and tickets while monitoring AI performance and viewing analytics such as ticket distribution, priority trends, workload, and customer satisfaction.

5. Developing Environment

The system will be developed on Windows 11 using **HTML, CSS, Bootstrap, and JavaScript** for the frontend and **Flask with Python** for the backend. **MySQL** will be used for storing user details, tickets, predictions, and resolved cases. NLP processing will use **NLTK** and **Sentence Transformers** for text analysis and semantic embeddings. Machine Learning models such as **Linear SVM** and **Random Forest** will be implemented using **Scikit-learn** for ticket classification and priority prediction. **Cosine Similarity** will be used for solution recommendation, while **Pandas, NumPy, Matplotlib** will support data processing and visualization. Development will be carried out using **Visual Studio Code**.

6. Conclusion

CompassIQ is an AI-powered customer support ticket intelligence system that automates ticket classification, prioritization, department routing, and solution recommendation using Natural Language Processing and Machine Learning. By analyzing customer complaints, predicting ticket attributes, and retrieving similar historical cases through semantic search, the proposed system significantly reduces manual effort while improving response accuracy and customer satisfaction.

The integration of Machine Learning, semantic similarity search, and modern web technologies demonstrates the effective application of Artificial Intelligence in customer support automation. CompassIQ provides a scalable, practical, and intelligent decision-support platform capable of enhancing operational efficiency in modern customer service environments.

7. References

- Sentence Transformers Documentation. Available: <https://www.sbert.net/>
- Scikit-learn: Machine Learning in Python. Available: <https://scikit-learn.org/>
- Flask Documentation. Available: <https://flask.palletsprojects.com/>
- Pandas Documentation. Available: <https://pandas.pydata.org/>
- NumPy Documentation. Available: <https://numpy.org/>
- NLTK Documentation. Available: <https://www.nltk.org/>

Abstract

Customer support organizations receive a large volume of customer complaints every day, making manual ticket categorization, prioritization, and routing increasingly challenging. Delays in assigning tickets to the appropriate departments and identifying suitable resolutions often lead to increased response times, inconsistent customer experiences, and higher operational costs. Traditional support systems primarily rely on manual processing, limiting their ability to efficiently manage growing ticket volumes. Recent advances in Natural Language Processing (NLP) and Machine Learning provide an effective solution by enabling intelligent analysis of customer complaints and automated decision-making throughout the ticket management process.

This project proposes **CompassIQ (Compass + Intelligence)**, an AI-powered customer support ticket routing and resolution system designed to automate ticket classification and support operations. The name **CompassIQ** combines the concept of a compass, representing intelligent guidance and routing, with artificial intelligence to highlight the system's ability to direct customer support requests toward the appropriate resolution path. The application analyzes customer complaints using NLP techniques to predict ticket categories, estimate priority levels, assign tickets to suitable support departments, retrieve semantically similar resolved tickets, and recommend the most appropriate solutions based on historical cases.

The proposed system integrates multiple Machine Learning techniques to improve prediction accuracy and support efficiency. Customer complaints are preprocessed using Natural Language Processing techniques and transformed into semantic embeddings using the **Sentence Transformer** model. A **Linear Support Vector Machine (SVM)** classifies support tickets into predefined categories, while a **Random Forest Classifier** predicts ticket priority. Semantic similarity search using cosine similarity retrieves the Top-3 most relevant historical tickets from the knowledge base, enabling reliable solution recommendations based on previously resolved cases. The application is developed using HTML, CSS, Bootstrap, JavaScript, Flask, MySQL, Scikit-learn, Sentence Transformers, NLTK, Pandas, and NumPy, providing an efficient, responsive, and intelligent web-based customer support platform. Additional features including role-based access control, prediction history, analytics dashboards, AI performance monitoring, and ticket management further enhance system usability and operational transparency.

The proposed solution assists customer support teams by reducing manual workload, improving ticket routing accuracy, shortening response times, and enabling faster issue resolution through intelligent recommendations. By integrating Natural Language Processing, Machine Learning, semantic search, and full-stack web technologies, CompassIQ demonstrates the practical application of Artificial Intelligence in modern customer support management while providing a scalable and industry-relevant solution suitable for enterprise environments.